THE MAXIMUM NUMBER OF ELEMENTS IN A MULTIPLICATIVE SIDON SET: LOWER BOUND

CHATGPT

Abstract. This is a blueprint written by ChatGPT in order to let Aristotle formalize a lower bound on the largest possible subset $A \subseteq \{1, 2, \ldots, n\}$ such that all products ab with $a, b \in A$ and $a \leq b$ are distinct. I first let ChatGPT write up Erdős' 1938 proof that shows that

$$|A| \geq \pi(n) + \frac{cn^{3/4}}{(\log n)^{3/2}}$$

is possible for some absolute constant c, and then asked it to optimize the constant c as much as possible, which eventually led to showing that any

$$c < \frac{2^{11/4}}{3^{3/4}} \approx 2.95$$

is admissible. This paper is entirely AI-generated, not independently verified, and not optimized for readability or ease of understanding. That being said, the proof is correct, as the Lean file that Aristotle produced can be found here: https://github.com/Woett/Lean-files/blob/main/ErdosProblem425Lower.lean.

Definition 1. *For a real number $x \geq 0$, let*

$$\pi(x) = \#\{p : p \text{ is prime and } p \leq x\}.$$

For $n \in \mathbb{N}$, let $[n] = \{1, 2, \ldots, n\}$.

Definition 2. *A finite set A of positive integers is called* product-Sidon *if for all $a, b, c, d \in A$,*

$$ab = cd$$

implies

$$(a = c \text{ and } b = d) \quad \text{or} \quad (a = d \text{ and } b = c).$$

Equivalently, the products ab with $a, b \in A$ and $a \leq b$ are all distinct.

Lemma 3 (Finite projective-plane block system)**.** *Let p be a prime, and let X be a finite set with*

$$|X| = p(p + 1) + 1.$$

Then there exist a finite index set I and subsets $C_i \subseteq X$ for $i \in I$ such that

$$|I| = p(p + 1) + 1,$$

$$|C_i| = p + 1 \qquad (i \in I),$$

$$|C_i \cap C_j| \leq 1 \qquad (i, j \in I,\ i \neq j),$$

and every point of X belongs to exactly $p + 1$ of the sets C_i.

Proof. It suffices to construct such a family for a set of cardinality p^2+p+1, since any two finite sets of the same cardinality are in bijection.

Let $F = \mathbb{F}_p$. Define the point set

$$Y = \{\mathrm{aff}(x, y) : x, y \in F\} \cup \{\inf(m) : m \in F\} \cup \{\infty\}.$$

Thus

$$|Y| = p^2 + p + 1.$$

Define the following subsets of Y. For $m, b \in F$, let

$$L_{m,b} = \{\mathrm{aff}(x, mx + b) : x \in F\} \cup \{\inf(m)\}.$$

For $a \in F$, let

$$V_a = \{\mathrm{aff}(a, y) : y \in F\} \cup \{\infty\}.$$

Finally, let

$$H = \{\inf(m) : m \in F\} \cup \{\infty\}.$$

Let

$$\mathcal{L} = \{L_{m,b} : m, b \in F\} \cup \{V_a : a \in F\} \cup \{H\}.$$

The members listed in this union are distinct. Indeed, the affine part of $L_{m,b}$ determines both m and b; the affine part of V_a determines a; no $L_{m,b}$ is equal to any V_a because the former contains $\inf(m)$ and the latter does not; and H is the only listed set with no affine points. Hence

$$|\mathcal{L}| = p^2 + p + 1.$$

Each member of $\mathcal{L}$ has cardinality $p + 1$.

It remains to check the intersections. If $(m, b) \neq (m', b')$, then two sets $L_{m,b}$ and $L_{m',b'}$ meet in exactly one point. Indeed, if $m = m'$ and $b \neq b'$, they meet only in $\inf(m)$. If $m \neq m'$, then the equation

$$mx + b = m'x + b'$$

has a unique solution $x \in F$, so the two sets meet in exactly one affine point.

For $L_{m,b}$ and V_a, the unique common point is

$$\mathrm{aff}(a, ma + b).$$

For $L_{m,b}$ and H, the unique common point is $\inf(m)$. For distinct $a, a' \in F$, the sets V_a and $V_{a'}$ meet only in ∞. Finally, V_a and H meet only in ∞.

Thus any two distinct members of $\mathcal{L}$ have intersection of cardinality 1, hence at most 1.

It remains to record the point incidences. An affine point $\mathrm{aff}(x, y)$ lies on the p lines $L_{m,y-mx}$, with $m \in F$, and on the vertical line V_x, hence on exactly $p + 1$ members of $\mathcal{L}$. The point $\inf(m)$ lies on the p lines $L_{m,b}$, with $b \in F$, and on H, hence on exactly $p + 1$ members of $\mathcal{L}$. Finally, ∞ lies on the p vertical lines V_a, with $a \in F$, and on H, hence on exactly $p + 1$ members of $\mathcal{L}$.

Transporting this family through a bijection $Y \to X$ gives the desired family on X. □

Lemma 4 (Layered block-product construction)**.** *Let $k \geq 1$ be an integer. For $1 \leq j \leq k$, let I_j be a finite set, let Q_j be a finite set of primes, and let*

$$R_j = \{r_{j,i} : i \in I_j\}$$

be a finite set of primes such that the map $i \mapsto r_{j,i}$ is injective for each fixed j. Let S be another finite set of primes. Assume that the sets of primes

$$S, \qquad Q_1, \dots, Q_k, \qquad R_1, \dots, R_k$$

are pairwise disjoint.

For each j and each $i \in I_j$, let $C_{j,i} \subseteq Q_j$. Assume that, for each fixed j,

$$|C_{j,i} \cap C_{j,i'}| \le 1 \qquad (i, i' \in I_j,\ i \ne i').$$

Define

$$A = S \cup \{r_{j,i} q : 1 \le j \le k,\ i \in I_j,\ q \in C_{j,i}\}.$$

Then A is product-Sidon. Moreover,

$$|A| = |S| + \sum_{j=1}^{k} \sum_{i \in I_j} |C_{j,i}|.$$

Proof. First observe that all displayed elements $r_{j,i}q$ are distinct. Indeed, if

$$r_{j,i}q = r_{j',i'}q',$$

with $q \in C_{j,i}$ and $q' \in C_{j',i'}$, then unique factorization and the pairwise disjointness assumptions imply $r_{j,i} = r_{j',i'}$ and $q = q'$. Pairwise disjointness of the sets R_j gives $j = j'$, and the injectivity of $i \mapsto r_{j,i}$ for this fixed j gives $i = i'$. Also, no element $r_{j,i}q$ belongs to S, since it is composite whereas every element of S is prime. Hence the cardinality formula follows.

It remains to prove the product-Sidon property. Suppose

$$ab = cd$$

with $a, b, c, d \in A$.

For a positive integer m, put

$$\nu_S(m) = \sum_{s \in S} v_s(m),$$

where $v_s(m)$ is the exponent of s in the prime factorization of m. If $x \in S$, then $\nu_S(x) = 1$, while if $x = r_{j,i}q$ for some j, i, q, then $\nu_S(x) = 0$. Therefore the number of elements among a, b which belong to S is equal to the number of elements among c, d which belong to S.

If both a and b belong to S, then both c and d belong to S, and unique factorization gives $\{a, b\} = \{c, d\}$ as unordered pairs.

If exactly one of a, b belongs to S, then, after swapping a, b and c, d if necessary, we may write

$$a = s \in S, \qquad b = r_{j,i}q, \qquad c = t \in S, \qquad d = r_{j',i'}q'.$$

The equality $sr_{j,i}q = tr_{j',i'}q'$ and unique factorization give $s = t$, $r_{j,i} = r_{j',i'}$, and $q = q'$. Hence $\{a, b\} = \{c, d\}$.

It remains to consider the case in which none of a, b, c, d belongs to S. Write

$$a = r_{j,i}q, \qquad b = r_{\ell,h}q', \qquad c = r_{j',i'}u, \qquad d = r_{\ell',h'}u',$$

where

$$q \in C_{j,i}, \quad q' \in C_{\ell,h}, \quad u \in C_{j',i'}, \quad u' \in C_{\ell',h'}.$$

The equality $ab = cd$ and unique factorization imply that the unordered pair of R-primes

$$\{r_{j,i}, r_{\ell,h}\}$$

is equal to

$$\{r_{j',i'}, r_{\ell',h'}\}.$$

After swapping c and d if necessary, we may assume

$$(j,i) = (j',i') \qquad \text{and} \qquad (\ell,h) = (\ell',h').$$

Then unique factorization gives

$$\{q,q'\} = \{u,u'\}$$

as multisets.

If $u = q$ and $u' = q'$, then $\{a,b\} = \{c,d\}$. The only other possibility is $u = q'$ and $u' = q$. If $(j,i) = (\ell,h)$, then again $\{a,b\} = \{c,d\}$. Suppose therefore that $(j,i) \neq (\ell,h)$. In the switched case we have

$$q \in C_{j,i} \cap C_{\ell,h}, \qquad q' \in C_{j,i} \cap C_{\ell,h}.$$

If $j \neq \ell$, this is impossible because $Q_j \cap Q_\ell = \varnothing$. If $j = \ell$, then the intersection condition gives $q = q'$. Hence also $u = u' = q$, and again $\{a,b\} = \{c,d\}$.

Thus every equality $ab = cd$ is trivial up to swapping the two factors, so A is product-Sidon. □

Lemma 5 (Near-square prime values)**.** *For every $\delta \in (0,1)$ there exists T_0 such that for every real $T \geq T_0$ there is a prime p satisfying*

$$(1-\delta)T \leq p(p+1) + 1 \leq T.$$

Proof. Let

$$y = \frac{-1 + \sqrt{4T-3}}{2}.$$

Then

$$y(y+1) + 1 = T.$$

Choose $\alpha \in (0,1)$ such that $\alpha^2 \geq 1 - \delta$. By the prime number theorem, for all sufficiently large y there is a prime p with

$$\alpha y \leq p \leq y.$$

Indeed,

$$\pi(y) - \pi(\alpha y) \sim \frac{(1-\alpha)y}{\log y} > 0.$$

For such a prime p, the upper bound $p \leq y$ gives

$$p(p+1) + 1 \leq y(y+1) + 1 = T.$$

The lower bound $p \geq \alpha y$ gives

$$p(p+1) + 1 \geq \alpha^2 y^2 + \alpha y + 1.$$

Since $y^2 + y + 1 = T$, we have

$$\alpha^2 y^2 + \alpha y + 1 = \alpha^2 T + (\alpha - \alpha^2)y + (1 - \alpha^2) \geq \alpha^2 T \geq (1-\delta)T.$$

This proves the lemma. □

Lemma 6 (Bounds for the cutoff recursion)**.** *Let*

$$x_0 = \frac{2}{3},$$

and define

$$x_{m+1} = 1 - \sqrt{\frac{1 - x_m}{2 - x_m}} \qquad (m \geq 0).$$

Then $0 < x_m < 1$ *for all* $m \geq 0$, $x_1 = 1/2$, *and* $x_m \leq 1/2$ *for all* $m \geq 1$.

Proof. Clearly $0 < x_0 < 1$. If $0 < x_m < 1$, then

$$0 < \frac{1 - x_m}{2 - x_m} < 1,$$

so $0 < x_{m+1} < 1$. This proves $0 < x_m < 1$ for all m by induction. Also,

$$x_1 = 1 - \sqrt{\frac{1/3}{4/3}} = \frac{1}{2}.$$

Finally, if $0 < x_m \leq 1/2$, then

$$\frac{1 - x_m}{2 - x_m} \geq \frac{1}{3},$$

and hence

$$x_{m+1} = 1 - \sqrt{\frac{1 - x_m}{2 - x_m}} \leq 1 - \frac{1}{\sqrt{3}} < \frac{1}{2}.$$

Since $x_1 = 1/2$, another induction gives $x_m \leq 1/2$ for all $m \geq 1$. □

Lemma 7 (The elementary cutoff functional identity)**.** *For* $0 \leq x < 1$, *put*

$$F(x) = \sqrt{2 - x}, \qquad f(x) = 1 - \sqrt{\frac{1 - x}{2 - x}}.$$

Then

$$F(x) = (1 - f(x))\sqrt{1 - x} + \sqrt{f(x)}F(f(x)).$$

Proof. Put

$$t = \sqrt{\frac{1 - x}{2 - x}}.$$

Then $f(x) = 1 - t$, so

$$1 - f(x) = t, \qquad 2 - f(x) = 1 + t.$$

Moreover,

$$1 - t^2 = 1 - \frac{1 - x}{2 - x} = \frac{1}{2 - x}.$$

Therefore

$$(1 - f(x))\sqrt{1 - x} = t\sqrt{1 - x} = \frac{1 - x}{\sqrt{2 - x}},$$

and

$$\sqrt{f(x)}F(f(x)) = \sqrt{1 - t}\sqrt{1 + t} = \sqrt{1 - t^2} = \frac{1}{\sqrt{2 - x}}.$$

Adding these two identities gives

$$(1 - f(x))\sqrt{1 - x} + \sqrt{f(x)}F(f(x)) = \frac{2 - x}{\sqrt{2 - x}} = \sqrt{2 - x} = F(x).$$

□

Lemma 8 (Cutoff constants approaching the limit)**.** *Put*

$$\Lambda = \frac{2^{11/4}}{3^{3/4}}.$$

For every $\gamma < \Lambda$ there exist an integer $k \geq 2$ and real numbers

$$0 = a_0 < a_1 < \cdots < a_k < 1$$

such that, with

$$d_j = a_j - a_{j-1} \qquad (1 \leq j \leq k),$$

and

$$b_j = \begin{cases} a_j^{-1} - a_{j+1}^{-1}, & 1 \leq j < k, \\ a_k^{-1} - a_k, & j = k, \end{cases}$$

one has

$$d_j \leq b_j \qquad (1 \leq j \leq k)$$

and

$$2 \sum_{j=1}^{k} d_j \sqrt{2b_j} > \gamma.$$

Proof. Let

$$x_0 = \frac{2}{3},$$

and define recursively

$$x_{m+1} = 1 - \sqrt{\frac{1 - x_m}{2 - x_m}} \qquad (m \geq 0).$$

By Lemma 6, we have $0 < x_m < 1$ for all m, $x_1 = 1/2$, and $x_m \leq 1/2$ for all $m \geq 1$.

For $k \geq 2$, define

$$a_j = \sqrt{x_0} \prod_{m=1}^{k-j} x_m \qquad (1 \leq j \leq k),$$

with the empty product interpreted as 1. Then

$$0 = a_0 < a_1 < \cdots < a_k < 1.$$

For $j < k$ we have $a_j \leq a_{k-1} = \sqrt{x_0} x_1$, hence $a_j^2 \leq 1/6$. Also, if $j < k$, then $a_j / a_{j+1} = x_{k-j} \leq 1/2$, so

$$b_j = \frac{1 - x_{k-j}}{a_j} \geq \frac{1}{2a_j} \geq a_j \geq d_j.$$

For the last layer,

$$d_k = a_k - a_{k-1} = \sqrt{x_0}(1 - x_1) = \frac{1}{2}\sqrt{\frac{2}{3}},$$

while

$$b_k = a_k^{-1} - a_k = \frac{1 - x_0}{\sqrt{x_0}} = \frac{1}{2}\sqrt{\frac{2}{3}}.$$

Thus $d_j \leq b_j$ for all j.

It remains to show that the corresponding constants tend to Λ. Put

$$F(x) = \sqrt{2 - x}, \qquad f(x) = 1 - \sqrt{\frac{1 - x}{2 - x}}.$$

By Lemma 7,

$$F(x) = (1 - f(x))\sqrt{1-x} + \sqrt{f(x)}F(f(x)) \qquad (0 \le x < 1).$$

Since $x_{m+1} = f(x_m)$, this identity may be iterated. Define

$$P_m = (x_1 x_2 \cdots x_m)^{1/2} \quad \text{and} \quad P_0 = 1.$$

For every $k \ge 2$, the iteration gives

$$F(x_0) = \sum_{m=0}^{k-2} P_m(1 - x_{m+1})\sqrt{1 - x_m} + P_{k-1}F(x_{k-1}).$$

Indeed, the term with index m is obtained after multiplying the identity at x_m by P_m, because

$$P_m\sqrt{x_{m+1}} = P_{m+1}.$$

We next compute the finite-layer sum in the same notation. Since

$$\frac{1}{2\sqrt{2}}\, 2\sum_{j=1}^{k} d_j\sqrt{2b_j} = \sum_{j=1}^{k} d_j\sqrt{b_j},$$

it is enough to rewrite the terms $d_j\sqrt{b_j}$. If $1 < j < k$ and $m = k - j$, then

$$a_j = x_0^{1/2} x_1 x_2 \cdots x_m, \qquad a_{j-1} = a_j x_{m+1},$$

and hence

$$d_j = a_j(1 - x_{m+1}), \qquad b_j = \frac{1 - x_m}{a_j}.$$

Thus

$$d_j\sqrt{b_j} = x_0^{1/4} P_m(1 - x_{m+1})\sqrt{1 - x_m}.$$

For the last layer, $j = k$, the same expression is the term $m = 0$:

$$d_k\sqrt{b_k} = x_0^{1/4}(1 - x_1)\sqrt{1 - x_0}.$$

For the first layer, $j = 1$, we have

$$d_1 = a_1, \qquad b_1 = \frac{1 - x_{k-1}}{a_1},$$

and therefore

$$d_1\sqrt{b_1} = x_0^{1/4} P_{k-1}\sqrt{1 - x_{k-1}}.$$

Combining the terms gives

$$\frac{1}{2\sqrt{2}}\, 2\sum_{j=1}^{k} d_j\sqrt{2b_j} = x_0^{1/4}\left(\sum_{m=0}^{k-2} P_m(1 - x_{m+1})\sqrt{1 - x_m} + P_{k-1}\sqrt{1 - x_{k-1}}\right).$$

Therefore the difference between

$$\frac{1}{2\sqrt{2}}\, 2\sum_{j=1}^{k} d_j\sqrt{2b_j}$$

and $x_0^{1/4}F(x_0)$ has absolute value at most

$$x_0^{1/4} P_{k-1}\sqrt{2}.$$

Since $x_m \le 1/2$ for $m \ge 1$, we have $P_{k-1} \to 0$. Hence

$$2\sum_{j=1}^{k} d_j\sqrt{2b_j} \longrightarrow 2\sqrt{2}\,x_0^{1/4}F(x_0).$$

Finally,

$$2\sqrt{2}\,x_0^{1/4}F(x_0) = 2\sqrt{2}\left(\frac{2}{3}\right)^{1/4}\sqrt{\frac{4}{3}} = \frac{2^{11/4}}{3^{3/4}} = \Lambda.$$

Thus, for every $\gamma < \Lambda$, choosing k sufficiently large gives the desired cutoffs. □

Lemma 9 (Layered prime supply)**.** *Fix an integer $k \ge 1$ and real numbers*

$$0 = a_0 < a_1 < \cdots < a_k < 1.$$

Put

$$d_j = a_j - a_{j-1} \qquad (1 \le j \le k),$$

and

$$b_j = \begin{cases} a_j^{-1} - a_{j+1}^{-1}, & 1 \le j < k, \\ a_k^{-1} - a_k, & j = k. \end{cases}$$

Assume that

$$d_j \le b_j \qquad (1 \le j \le k),$$

and put

$$L = 2\sum_{j=1}^{k} d_j\sqrt{2b_j}.$$

Then, for every $\eta > 0$, there exists N_0 such that for every integer $n \ge N_0$ there are primes $p_1, \dots, p_k$ and integers $v_1, \dots, v_k$ with the following property. If

$$\rho_j = p_j(p_j + 1) + 1,$$

then

$$v_j \le \pi(a_j\sqrt{n}) - \pi(a_{j-1}\sqrt{n}) \qquad (1 \le j \le k),$$
$$v_j \le \rho_j \qquad (1 \le j \le k),$$
$$\rho_j \le \pi(a_j^{-1}\sqrt{n}) - \pi(a_{j+1}^{-1}\sqrt{n}) \qquad (1 \le j < k),$$
$$\rho_k \le \pi(a_k^{-1}\sqrt{n}) - \pi(a_k\sqrt{n}),$$

and

$$\sum_{j=1}^{k} v_j(p_j + 1) \ge (L - \eta)\frac{n^{3/4}}{(\log n)^{3/2}}.$$

Proof. Put

$$X_n = \frac{\sqrt{n}}{\log n}.$$

Choose $\varepsilon > 0$ sufficiently small that

$$2d_j - 2\varepsilon > 0, \qquad 2b_j - \frac{\varepsilon}{2} > 0 \qquad (1 \le j \le k),$$

and

$$\sum_{j=1}^{k}(2d_j - 2\varepsilon)\sqrt{2b_j - \frac{\varepsilon}{2}} > L - \eta.$$

By the prime number theorem, for every fixed $\alpha > 0$,

$$\pi(\alpha\sqrt{n}) \sim 2\alpha X_n.$$

For $j \geq 2$, this gives

$$\pi(a_j\sqrt{n}) - \pi(a_{j-1}\sqrt{n}) \sim 2d_j X_n.$$

For $j = 1$, the same conclusion follows separately from $a_0 = 0$:

$$\pi(a_1\sqrt{n}) - \pi(a_0\sqrt{n}) = \pi(a_1\sqrt{n}) \sim 2a_1 X_n = 2d_1 X_n.$$

Hence, for all sufficiently large n,

$$\pi(a_j\sqrt{n}) - \pi(a_{j-1}\sqrt{n}) \geq (2d_j - \frac{\varepsilon}{2})X_n \qquad (1 \leq j \leq k).$$

Similarly, for $1 \leq j < k$,

$$R_{j,n} := \pi(a_j^{-1}\sqrt{n}) - \pi(a_{j+1}^{-1}\sqrt{n}) \sim 2b_j X_n,$$

and

$$R_{k,n} := \pi(a_k^{-1}\sqrt{n}) - \pi(a_k\sqrt{n}) \sim 2b_k X_n.$$

Therefore, after increasing the lower bound on n if necessary,

$$R_{j,n} \geq (2b_j - \frac{\varepsilon}{4})X_n \qquad (1 \leq j \leq k).$$

In particular, because $2b_j - \varepsilon/4 > 0$ and $X_n \to \infty$, each $R_{j,n}$ tends to infinity as $n \to \infty$. Increasing the lower bound on n if necessary, we may also assume that

$$\lfloor (2d_j - \varepsilon)X_n \rfloor \geq (2d_j - 2\varepsilon)X_n \qquad (1 \leq j \leq k).$$

Choose $\delta \in (0, 1)$ so small that

$$(1 - \delta)\left(2b_j - \frac{\varepsilon}{4}\right) \geq 2b_j - \frac{\varepsilon}{2} \qquad (1 \leq j \leq k).$$

Let T_0 be the threshold supplied by Lemma 5 for this value of δ. Since $R_{j,n} \to \infty$ for each fixed j, after increasing N_0 once more we have $R_{j,n} \geq T_0$ for all $n \geq N_0$ and all j. Hence, for every $n \geq N_0$ and every j, Lemma 5 gives a prime p_j satisfying

$$(1 - \delta)R_{j,n} \leq p_j(p_j + 1) + 1 \leq R_{j,n}.$$

Put

$$\rho_j = p_j(p_j + 1) + 1, \qquad v_j = \lfloor (2d_j - \varepsilon)X_n \rfloor.$$

Then

$$v_j \leq \pi(a_j\sqrt{n}) - \pi(a_{j-1}\sqrt{n}),$$

and the upper bound $\rho_j \leq R_{j,n}$ gives the asserted upper bounds for ρ_j in the appropriate prime intervals.

Moreover,

$$\rho_j \geq (2b_j - \frac{\varepsilon}{2})X_n.$$

Since $d_j \leq b_j$,

$$v_j \leq (2d_j - \varepsilon)X_n \leq (2b_j - \varepsilon)X_n < (2b_j - \frac{\varepsilon}{2})X_n \leq \rho_j.$$

Finally, since $\rho_j = p_j(p_j + 1) + 1 < (p_j + 1)^2$, we have $p_j + 1 > \sqrt{\rho_j}$. Therefore

$$\sum_{j=1}^{k} v_j(p_j + 1) > \sum_{j=1}^{k} v_j\sqrt{\rho_j} \geq \sum_{j=1}^{k}(2d_j - 2\varepsilon)\sqrt{2b_j - \frac{\varepsilon}{2}}\, X_n^{3/2}.$$

By the choice of ε, this is at least

$$(L-\eta)\frac{n^{3/4}}{(\log n)^{3/2}}.$$

This proves the lemma. □

Theorem 10. *For every constant c with*

$$0 < c < \frac{2^{11/4}}{3^{3/4}},$$

there exists N such that for every integer $n \geq N$ there exists a product-Sidon set

$$A \subseteq [n]$$

with

$$|A| \geq \pi(n) + c\frac{n^{3/4}}{(\log n)^{3/2}}.$$

Proof. Put

$$\Lambda = \frac{2^{11/4}}{3^{3/4}}.$$

Fix c with $0 < c < \Lambda$. By Lemma 8, choose $k \geq 2$ and cutoffs

$$0 = a_0 < a_1 < \cdots < a_k < 1$$

such that, with d_j and b_j as in Lemma 9,

$$d_j \leq b_j \qquad (1 \leq j \leq k)$$

and

$$L := 2\sum_{j=1}^{k} d_j\sqrt{2b_j} > c.$$

Choose $\eta > 0$ such that

$$L - \eta > c.$$

Choose N large enough so that Lemma 9 applies for every $n \geq N$, so that $a_1^{-1}\sqrt{n} \leq n$ for every $n \geq N$, and also

$$\pi(a_1^{-1}\sqrt{n}) \leq (L-\eta-c)\frac{n^{3/4}}{(\log n)^{3/2}}$$

for every $n \geq N$. Such an N exists because a_1 is fixed and

$$\pi(a_1^{-1}\sqrt{n}) = O\left(\frac{\sqrt{n}}{\log n}\right) = o\left(\frac{n^{3/4}}{(\log n)^{3/2}}\right).$$

Fix $n \geq N$. By Lemma 9, choose primes $p_1, \ldots, p_k$ and integers $v_1, \ldots, v_k$ satisfying the conclusions of that lemma, and put

$$\rho_j = p_j(p_j+1)+1.$$

For each j, choose a set Q_j of v_j primes in the interval

$$(a_{j-1}\sqrt{n},\ a_j\sqrt{n}].$$

This is possible because

$$v_j \leq \pi(a_j\sqrt{n}) - \pi(a_{j-1}\sqrt{n}).$$

Choose a finite set X_j such that

$$Q_j \subseteq X_j \qquad \text{and} \qquad |X_j| = \rho_j.$$

This is possible because $v_j \le \rho_j$.

By Lemma 3, for each j there exist a finite index set I_j and subsets $D_{j,i} \subseteq X_j$ for $i \in I_j$ such that

$$|I_j| = \rho_j,$$
$$|D_{j,i}| = p_j + 1 \qquad (i \in I_j),$$
$$|D_{j,i} \cap D_{j,i'}| \le 1 \qquad (i, i' \in I_j,\ i \ne i'),$$

and every point of X_j belongs to exactly $p_j + 1$ of the sets $D_{j,i}$. For $i \in I_j$, define

$$C_{j,i} = D_{j,i} \cap Q_j.$$

Then $C_{j,i} \subseteq Q_j$, and for fixed j,

$$|C_{j,i} \cap C_{j,i'}| \le 1 \qquad (i, i' \in I_j,\ i \ne i').$$

Moreover,

$$\sum_{i \in I_j} |C_{j,i}| = \sum_{i \in I_j} |D_{j,i} \cap Q_j| = (p_j + 1)\,|Q_j| = v_j(p_j + 1),$$

because every point of Q_j belongs to exactly $p_j + 1$ of the sets $D_{j,i}$.

For $1 \le j < k$, choose distinct primes

$$r_{j,i} \in (a_{j+1}^{-1}\sqrt{n},\ a_j^{-1}\sqrt{n}] \qquad (i \in I_j).$$

This is possible because

$$|I_j| = \rho_j \le \pi(a_j^{-1}\sqrt{n}) - \pi(a_{j+1}^{-1}\sqrt{n}).$$

For $j = k$, choose distinct primes

$$r_{k,i} \in (a_k\sqrt{n},\ a_k^{-1}\sqrt{n}] \qquad (i \in I_k),$$

which is possible because

$$|I_k| = \rho_k \le \pi(a_k^{-1}\sqrt{n}) - \pi(a_k\sqrt{n}).$$

Let

$$S = \{s : s \text{ is prime and } a_1^{-1}\sqrt{n} < s \le n\}.$$

Define

$$A = S \cup \{r_{j,i}q : 1 \le j \le k,\ i \in I_j,\ q \in C_{j,i}\}.$$

We first check that $A \subseteq [n]$. If $s \in S$, then $s \le n$. If $q \in C_{j,i}$ and $j < k$, then

$$q \le a_j\sqrt{n} \qquad \text{and} \qquad r_{j,i} \le a_j^{-1}\sqrt{n},$$

so $r_{j,i}q \le n$. If $q \in C_{k,i}$, then

$$q \le a_k\sqrt{n} \qquad \text{and} \qquad r_{k,i} \le a_k^{-1}\sqrt{n},$$

so again $r_{k,i}q \le n$.

The sets of primes

$$S, \qquad Q_1, \ldots, Q_k, \qquad \{r_{j,i} : i \in I_j\} \quad (1 \le j \le k)$$

are pairwise disjoint. The sets Q_j lie in the pairwise disjoint intervals

$$(a_{j-1}\sqrt{n},\ a_j\sqrt{n}] \qquad (1 \le j \le k).$$

The primes $r_{k,i}$ lie in

$$(a_k\sqrt{n}, a_k^{-1}\sqrt{n}],$$

which is disjoint from every Q_j because every prime in every Q_j is at most $a_k\sqrt{n}$, while the left endpoint for the $r_{k,i}$ interval is open. For $j < k$, the primes $r_{j,i}$ lie in

$$(a_{j+1}^{-1}\sqrt{n},\ a_j^{-1}\sqrt{n}].$$

These intervals are pairwise disjoint as j varies. They are also disjoint from the $r_{k,i}$ interval: the interval for $j = k-1$ begins at the open endpoint $a_k^{-1}\sqrt{n}$, and all earlier such intervals lie still farther to the right. Finally, S consists of primes larger than $a_1^{-1}\sqrt{n}$, whereas every $r_{j,i}$ is at most $a_1^{-1}\sqrt{n}$ and every prime in every Q_j is at most $a_k\sqrt{n} < a_1^{-1}\sqrt{n}$. Therefore Lemma 4 applies and shows that A is product-Sidon.

By the cardinality formula in Lemma 4,

$$|A| = |S| + \sum_{j=1}^{k}\sum_{i\in I_j}|C_{j,i}| = \pi(n) - \pi(a_1^{-1}\sqrt{n}) + \sum_{j=1}^{k} v_j(p_j+1).$$

Using Lemma 9, we get

$$|A| \geq \pi(n) - \pi(a_1^{-1}\sqrt{n}) + (L-\eta)\frac{n^{3/4}}{(\log n)^{3/2}}.$$

By the choice of N,

$$\pi(a_1^{-1}\sqrt{n}) \leq (L-\eta-c)\frac{n^{3/4}}{(\log n)^{3/2}}.$$

Hence

$$|A| \geq \pi(n) + c\frac{n^{3/4}}{(\log n)^{3/2}}.$$

This proves the theorem. □